

ACA Lab-8: Shared Memory Multiprogramming-III: Barriers

AIM: Programs using barrier

EXERCISES:

1. The header.h file created in LAB 5 includes barrier_init() and barrier() function. =>

```
#include <sys/wait.h>
#include <unistd.h>
#include <sys/shm.h>
#include <stdlib.h>
#include <sys/sem.h>
#include <stdio.h>

int process_fork(int nproc){
    int j;
    for(j=1;j<nproc;j++)
        if(fork()==0) return j;
    return 0;
}

void process_join(int nproc,int id){
    int i;
    if(id==0)
        for(i=1;i<nproc;i++)
            wait(0);
    else exit(0);
}

char* shared(int size,int *shmid){
    *shmid=shmget(IPC_PRIVATE,size,0666|IPC_CREAT);
    return (shmat(*shmid,0,0));
}

void spin_lock_init (int *lock, int *condition){
    int control;
    *lock= semget(IPC_PRIVATE,1,0666|IPC_CREAT);
    if(*condition==1) control=0;
    else control=1;
```

```

semctl(*lock,0,SETVAL,control);
}

void spin_lock (int *lock){
    struct sembuf operations;
    operations.sem_num =0;
    operations.sem_op=-1;
    operations.sem_flg =0;
    semop (*lock,&operations,1);
}

void spin_unlock(int *lock){
    struct sembuf operations;
    operations.sem_num=0;
    operations.sem_op=1;
    operations.sem_flg=0;
    semop(*lock,&operations,1);
}

void barrier_init (int *bar1,int bnum){
    int c=0;
    *(bar1+0)=bnum;
    *(bar1+1)=0;
    *(bar1+2)=0;
    spin_lock_init((bar1+3),&c);
}

void barrier (int *bar){
    int incr =0;
    while(1){
        spin_lock((bar+3));
        if (incr==0 && *(bar+2)>0){
            spin_unlock((bar+3)); continue;
        }
        if (incr==0){
            *(bar+1)=*(bar+1)+1;
            incr =1;
        }
    }
}

```

```

if(*(bar+1) < *(bar+0) && (*(bar+2) ==0)){
spin_unlock((bar+3));
continue;
}
else {
if(*(bar+2)==0){
*(bar+2)=*(bar+0)-1;
*(bar+1)=0;
spin_unlock((bar+3));
return;

}
else {
*(bar+2)= *(bar+2)-1;
spin_unlock((bar+3));
return;
}
}
}
spin_unlock((bar+3));
printf("\n Error in barrier");
return;
}

```

2. WAP to Calculate standard deviation with help of barrier

=>

```

#include "header.h"
#include <math.h>
int main() {
int shmid, id ,i, locked=1,unlocked=0,*lock, nproc,*bar;
float *average,*deviation, sum=0.0,a[20];
lock=(int *)shared (sizeof (int), &shmid);
deviation=(float *)shared (sizeof (float), &shmid);
bar=(int *)shared (sizeof (int)*4,&shmid);

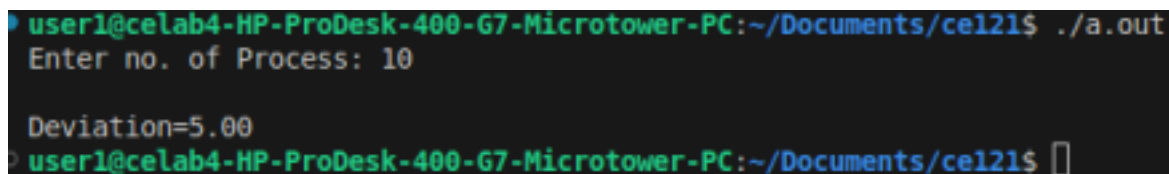
```

```

average=(float *)shared (sizeof (float), &shmid);
*average=0.0;
*deviation=0.0;
printf("Enter no. of Process: ");
scanf("%d", &nproc);
for (i=0;i<20;i++)
a[i]=i;
spin_lock_init (lock, &unlocked);
barrier_init (bar, nproc);
printf("\n");
id=process_fork (nproc);
for (i=id; i<20;i+=nproc)
sum= sum+ a[i];
spin_lock (lock); //average
*average=*average+sum/20;
spin_unlock (lock);
barrier(bar); //barrier used
sum=0.0;
for (i=id; i<20;i+=nproc)
sum=sum + fabs (a[i]-*average);
spin_lock(lock);
*deviation=*deviation+sum/20;
spin_unlock (lock);
process_join (nproc, id);
printf("Deviation=%.2f\n",*deviation);
return(0);
}

```

Output :



```

user1@celab4-HP-ProDesk-400-G7-Microtower-PC:~/Documents/cel21$ ./a.out
Enter no. of Process: 10

Deviation=5.00
user1@celab4-HP-ProDesk-400-G7-Microtower-PC:~/Documents/cel21$ 

```

3. Write a parallel program to implement Histogram with a barrier.

=>

```
#include "header.h"
#include <stdio.h>
#define arrSize 15
void find_max_min(int a[],int id, int *sm, int *sn, int *lock1,int *bar,int nproc){
int i, amax, amin;
if (id==0){
*sm=a[0];
*sn=a[0];
}
barrier(bar);
amax=a[0];
amin=a[0];
for (i=id ;i<10;i+=nproc){
if (amax <a[i])
amax=a[i];
if (amin > a[i])
amin=a[i];
}
spin_lock(lock1);
if(*sm < amax)
*sm=amax;
if(*sn > amin)
*sn=amin;
spin_unlock(lock1);
barrier (bar);
}
int main() {
int a[arrSize],NoOfBins=5,binsize,*histogram,*lock1,*lock2, unlocked=0,locked=1,*bar,
shmidbar ,shmidlock1, shmidlock2,shmidhist,shmidmax,shmidmin,id, nproc,bin,*amin, *amax,
i;
amax =(int *)shared (sizeof (int), &shmidmax);
amin =(int *)shared (sizeof (int), &shmidmin);
bar=(int *)shared (sizeof (int)*4, &shmidbar);
```

```

lock1=(int *)shared (sizeof (int), &shmidlock1);
lock2=(int*) shared (sizeof (int)*NoOfBins, &shmidlock2);
barrier_init (bar, nproc);
for (i=0;i<NoOfBins; i++)
spin_lock_init((lock2+i),&unlocked);
spin_lock_init(lock1,&unlocked);
for (i=0;i< arrSize;i++){
printf("enter a[%d] :",i);
scanf("%d",& a[i]);
}
nproc = NoOfBins;
id=process_fork (nproc);
find_max_min(a,id,amax,amin,lock1,bar,nproc);
binsize=(amax-amin) / NoOfBins;
histogram=(int*)shared (sizeof(int)* NoOfBins,&shmidhist);
printf("\n Bin Size : %d", binsize);
printf("\n No. Of Bins : %d \n", NoOfBins);
for (i=0;i<NoOfBins; i++)
*(histogram+ i)=0;
printf("\n");
for( i=id; i<arrSize; i=i+nproc){
bin=abs (a[i] - *amin)/binsize;
if (bin>=NoOfBins)
bin=NoOfBins - 1;
printf ("Process( %d ):Number [%d] is %d \t Bin: %d \n", id, i, a[i] ,bin);
spin_lock (lock2+ bin);
*(histogram+ bin)+=1;
spin_unlock (lock2 +bin);
}
printf("\n");
process_join (NoOfBins, id);
for (i=0;i<NoOfBins; i++)
printf ("No Of Items in Bin (%d) is %d \n", i,*(histogram+ i));
return 0;

```

}

Output :

```
user1@celab5-Veriton-M200-P500:~/Downloads/cel2932_cn/RS-232$ ./a.  
Enter No Of Bins : 5  
enter a[0] : 0  
enter a[1] : 11  
enter a[2] : 12  
enter a[3] : 19  
enter a[4] : 22  
enter a[5] : 23  
enter a[6] : 24  
enter a[7] : 25  
enter a[8] : 27  
enter a[9] : 29  
enter a[10] : 35  
enter a[11] : 36  
enter a[12] : 37  
enter a[13] : 45  
enter a[14] : 50  
  
Bin Size : 10  
No. Of Bins : 5  
  
Process(1):Number[1] is 11      Bin: 1  
Process(1):Number[6] is 24      Bin: 2  
Process(1):Number[11] is 36     Bin: 3  
  
Process(2):Number[2] is 12      Bin: 1  
Process(2):Number[7] is 25      Bin: 2  
Process(2):Number[12] is 37     Bin: 3  
  
Process(3):Number[3] is 19      Bin: 1  
Process(3):Number[8] is 27      Bin: 2  
Process(3):Number[13] is 45     Bin: 4  
  
Process(0):Number[0] is 0       Bin: 0  
Process(0):Number[5] is 23      Bin: 2  
Process(0):Number[10] is 35     Bin: 3  
  
Process(4):Number[4] is 22      Bin: 2  
Process(4):Number[9] is 29      Bin: 2  
Process(4):Number[14] is 50     Bin: 4  
  
No Of Items in Bin (0) is 1  
No Of Items in Bin (1) is 3  
No Of Items in Bin (2) is 6  
No Of Items in Bin (3) is 3  
No Of Items in Bin (4) is 2  
user1@celab5-Veriton-M200-P500:~/Downloads/cel2932_cn/RS-232$
```